

Proof that $\det \circ \exp = \exp \circ \text{tr}$

Ingredient. *Triangular matrices are closed under multiplication.* Let A and B be upper triangular matrices, so that $A_{ij} = B_{ij} = 0$ for $i > j$. From

$$(AB)_{ij} = \sum_k A_{ik} B_{kj} = \sum_k \begin{cases} A_{ik} B_{kj} & \text{if } i \leq k \leq j \\ 0 & \text{otherwise} \end{cases}$$

it follows that AB is also upper triangular. In particular, $(AB)_{ii} = A_{ii} B_{ii}$.

Ingredient. *Only diagonal elements of triangular matrices affect the trace.*

$$\text{tr}(AB) = \sum_k (AB)_{kk} = \sum_k A_{kk} B_{kk} = \text{diag}(A) \cdot \text{diag}(B)$$

Ingredient. *Only diagonal elements of triangular matrices affect the determinant.* If A is triangular, then

$$\det(A) = \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^n A_{i\sigma(i)} = \prod_{i=1}^n A_{ii}$$

because all the permutations σ except the identity have some $1 \leq k \leq n$ such that $\sigma(k) < k$.

Ingredient. *Any square matrix A can be put in Jordan normal form $A = PJP^{-1}$, where J is upper triangular.*

$$\begin{aligned} \det(\exp(A)) &= \det(\exp(PJP^{-1})) \\ &= \det(P \exp(J) P^{-1}) \\ &= \det(P) \det(\exp(J)) \det(P^{-1}) \\ &= \det(\exp(J)) \\ &= \prod_{i=1}^n \exp(J)_{ii} \\ &= \prod_{i=1}^n \sum_{n=0}^{\infty} \frac{1}{n!} (J^n)_{ii} \\ &= \prod_{i=1}^n \sum_{n=0}^{\infty} \frac{1}{n!} (J_{ii})^n \\ &= \prod_{i=1}^n \exp(J_{ii}) \\ &= \exp\left(\sum_{i=1}^n J_{ii}\right) \\ &= \exp(\text{tr}(J)) \\ &= \exp(\text{tr}(PP^{-1}J)) \\ &= \exp(\text{tr}(PJP^{-1})) \\ &= \exp(\text{tr}(A)) \end{aligned}$$